



TAKE-HOME ASSIGNMENT

HVAC Data Analysis & Control Strategy Backtest

ALTO BUILDING · FLOOR 8 DATASET



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1. About This Assignment

Thank you for your time so far. This take-home is the practical half of our process: we want to see how you turn a real, imperfect building dataset into an insight, a control decision, and a quantified result — in code, end to end.

You will receive three months of 15-minute telemetry from one floor of an office building that AltoTech operates. The floor is served by two air-handling units with variable-speed fans, 53 VAV boxes, fan coil units, exhaust fans, power meters, and wireless indoor air quality sensors. The data is exactly as it came out of the building management system. Nothing has been cleaned.

Item	Value
What we assess	Data handling, engineering judgement, choice and justification of a control strategy, rigour and honesty of the backtest, code quality, and how clearly you communicate the result
Expected effort	8–12 hours of focused work. Depth on a few things beats breadth on everything
Time allowed	7 calendar days from the day you receive the dataset
Deliverables	One GitHub repository and one PDF report — see Section 6
Follow-up	A follow-up session to discuss your submission
Tools	Any language and any library. AI assistants are allowed

There is no single right answer. We are more interested in how you reason under uncertainty than in the size of the number you arrive at. A modest, well-defended saving with clear assumptions scores higher than a large one that cannot be traced back to the data.



2. The Dataset

Item	Value
File	alto_building_floor8_15min_2026-05-18_to_2026-08-23.csv (28 MB)
Period	18 May 2026 12:00 to 23 Aug 2026 10:00 local (Asia/Bangkok, UTC+7)
Columns	timestamp_local (bucket start), timestamp_utc , then 663 columns, one per point, named <device_id>__<datapoint>
Values	9,305 rows on a complete 15-minute grid. Each cell is the aggregate of the underlying 1-minute samples; for on/off points, the fraction of the slot the point was on. Empty cells mean no data was recorded
Units	Not documented. Infer them from the values and the physics

Equipment on the floor

Model	Devices	Datapoints
ahu	ahu_b8_1 , ahu_b8_2	Supply/return air temperature, static pressure, VSD frequency, CO ₂ , return air humidity and water delta-T, each with setpoints; power, current, voltage; cooling valve position / flow / energy; chilled water supply/return temperature; fresh air damper; cooling rate; status, control modes, alarms
vav	53 boxes: vav_8_1_1...28 , vav_8_2_1...28	Room temperature, room setpoint read/write, damper position, air flow rate, minimum and maximum air flow setpoints
indoor_air_quality	floor_8_zone_1_iaq_1...13	Temperature, humidity, battery, RSSI, online status. Sensors 1–4 also CO ₂ , TVOC, PM2.5, PM10, pressure, light level, PIR occupancy
power_meter	db_b8 — the floor's main distribution board pp_b8_1 , pp_b8_2 — sub-panels fed from it power_meter_1 , power_meter_2 — two further meters	Active, reactive and apparent power, power factor, per-phase current and voltage, cumulative energy, frequency
fcu , exhaust_fan	fcu_b8_1...14 ; ex_b8_1 , ex_b8_2	Status, alarm; exhaust fans also status write, auto/manual mode, trip alarm
co2_sensor , weather station	co2_duct_sensor_1 , _2 ; outdoor_weather_station	Duct CO ₂ concentration; outdoor dry-bulb, wet-bulb, humidity
modbus_gateway	1	Connection status, devices online / total, error count, response time

Naming. b8 and a leading 8 both mean floor 8. vav_8_1_* boxes are on AHU 1's supply duct and vav_8_2_* on AHU 2's. A _read / _write pair is the feedback and the command of the same point.



3. Task A — Exploratory Analysis, Problems & Opportunities

Before controlling anything, understand what the floor is doing today. We want a view of the system's behaviour that a facilities manager could act on, backed by the data.

1 Load and validate

Load the CSV, establish the time index, and characterise coverage. Decide, and state, how you treat missing data. Do not assume a value is trustworthy because it is present.

2 Describe the operating pattern

Operating hours, weekday versus weekend behaviour, how the AHUs are staged, how fan speed and static pressure respond to zone demand, how the VAV boxes behave relative to their setpoints, and how cooling delivered relates to outdoor conditions and occupancy.

3 Identify problems

Anything that is broken, mis-tuned, hunting, saturated, fighting another loop, or running when it should not. Rank by impact on energy or comfort and support each one with a figure or a number.

4 Identify opportunities

Where is energy being spent without buying comfort or air quality? Which of these could a supervisory controller act on using only the points available in this dataset? This list is your shortlist for Task B.

What good looks like

- A short data-quality section that says what you trust, what you do not, and why.
- Three to six well-chosen figures, each with a one-sentence takeaway. Not a wall of plots.
- Problems and opportunities stated in engineering terms with a rough size attached.
- Awareness that a building dataset reflects how the plant was actually run, including operator interventions, not just how it was designed to run.

Which loads sit behind each meter is not documented. Section 2 gives the meter hierarchy. Verify it against the data and decide which meter is the right baseline for your strategy before you rely on it. Getting this wrong quietly invalidates everything in Task B.



4. Task B — Control Strategy & Backtest

Pick a supervisory control strategy that addresses the opportunities you found in Task A, implement it in code, and backtest it against the historical data.

Choose your strategy

Pick one from the menu, combine several, or propose your own. Explain why it fits this floor best.

Strategy	Idea	Lever → Effect
Static pressure reset	Reset the duct static pressure setpoint from VAV damper demand (e.g. trim-and-respond) so the fan only makes the pressure the worst zone needs	Static pressure setpoint → VSD frequency, fan power
Supply air temperature reset	Raise the supply air setpoint when zone cooling demand is low, subject to humidity and comfort limits	SAT setpoint → cooling valve, cooling energy
Chilled water delta-T management	Keep the coil's water delta-T near design by managing valve position and flow, reducing over-pumping	Valve / flow → delta-T, cooling rate
Zone setpoint and airflow optimisation	Reset room setpoints or VAV airflow limits inside a comfort band using occupancy and IAQ signals; optimise start, stop and unoccupied periods	Room setpoint / airflow limits → zone demand, fan and cooling energy
Your own	Anything the dataset supports. State the control input, the measured effect, and the constraint it must respect	—

Implement it

Write the controller as a function of the data available at each timestep: measurements up to time t in, the command at t out. Make tuning parameters explicit and configurable, and make it obvious what it does when data is missing or a sensor reads nonsense.

Backtest it

Replay the controller over the full period and compare its commands with what the building actually did, then translate the difference in control input into an estimated energy difference using a stated relationship, e.g. fan affinity laws or the coil load relationship.

We know a replay is not a closed-loop simulation. The building never responded to your controller, so its true reaction is not in the data. Comparing on the control input with a transparent effect model is the best available here. State what the estimate assumes and give a range where the assumptions are weak.

- **Define the baseline** precisely: period, columns, meter, and why.
- **State a comfort constraint** and show the controller respects it. A zone temperature band is the minimum. Thermal comfort is more than temperature, so a comfort index such as PMV, computed from the available points with stated assumptions, is the stronger test. Savings bought with discomfort do not count.
- **Test sensitivity** to your tuning parameters and effect-model assumptions.
- **Benchmark** first against what the building actually commanded in the dataset. Beyond that, be creative: a simpler alternative such as a fixed setpoint change shows what your strategy adds.



5. Task C — Results & Recommendation

Bring it together into something a decision-maker can act on. Quantify, qualify, and say what you would do next.

Quantify the opportunity

There is no fixed template for this. We want your own way of putting a number on what your strategy would achieve, and your reasoning for it. Cover, in whatever form fits your work:

- **Why it should work** on this floor specifically, with the evidence from Task A that supports it.
- **What is expected to change:** which quantities go down, or get better, by roughly how much, and over what period. Energy is the obvious one, but comfort, air quality, equipment runtime and stability all count.
- **Why those numbers:** the assumptions and the relationship between the control input you change and the effect you claim, with a range where the assumptions are weak.
- **How you would measure it** once the strategy runs on the real building: which points, what baseline, over what period, and what would tell you it is not working.

Qualify the result

- Which assumptions drive the number most, and what happens to it if they are wrong.
- Where the data quality limited what you could conclude.
- Periods where the controller would have behaved badly, and what guarded against that.

Recommend

Close with a one-page recommendation: should AltoTech deploy this strategy on this floor? If yes, what would you need before going live — sensor fixes, interlocks, override and rollback behaviour, tuning period, monitoring? If no, what would change your mind? We are looking for the judgement of someone who will be responsible for a controller running on a real building.



6. Deliverables, Timeline & Ground Rules

Deliverables

Deliverable	Requirements
1. GitHub repository	Private repository, share access with your AltoTech contact. Contains all code for Tasks A–C, a <code>README</code> with setup and run instructions, a dependency file, and a way to reproduce every figure and number in your report from the CSV. Notebooks are fine; a script that runs end to end is better. Do not commit the dataset.
2. PDF report	At most 10 pages including figures. Structure it around Tasks A, B and C. Written for an engineering manager who has not seen the data. Every number should be traceable to a place in the repository.

Timeline

- Day 1 — You receive this document and the dataset**
Confirm receipt and that you can open the file. Ask clarifying questions by email at any time; we answer within one working day.
- Day 7, 23:59 Bangkok time — Submission**
Email the repository link and the PDF to your AltoTech contact. Commits after this time are not considered.
- Within the following week — Follow-up session**
We schedule a session to discuss your submission and the results, and to talk about how we work.

Ground rules

- **AI assistants are allowed.** Use whatever helps you work well. You are responsible for every line and every claim in what you submit.
- **The data is confidential.** Use it for this assignment only, do not share or publish it, do not attempt to identify the building, and delete your copy once the process is complete.
- **Talk to us.** Reach out to the team by email at any time, in the same thread. Treat this as working with colleagues on a real problem, not as an exam.
- **Individual work.** Discussing ideas with others is fine; the code and the report must be your own.
- **Life happens.** If something gets in the way of the deadline, tell us early and we will find a solution together.

We look forward to seeing what you find. Good luck.